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Dialogue relation extraction based on Multi-Perspective Contextual Interaction Attention Network

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ABSTRACT

Dialogue relation extraction aims to identify semantic relations between entity pairs from dialogue text. Multi-party dialogues present unique challenges due to dynamic speaker interactions and cross-turn dependencies, which existing methods fail to adequately model. We propose the Multi-Perspective Contextual Interaction Attention Network (**MPCIA-Net**) to address these limitations through two key components: (1) The Dual-stream Context Disentanglement Module (**DCDM**), which decouples context representations from the base encoder through parallel feature streams and integrates fine-grained contextual features; (2) The Dual-Attention Fusion Module (**DAF**), which processes features from dual perspectives—speaker perception for pronoun resolution through dialogue participant identity modelling, and utterance perception for capturing long-range cross-turn dependencies. The representations from both channels are merged through an adaptive attention mechanism for final relation prediction. Experimental results demonstrate that MPCIA-Net achieves an F_1 score of 68.70% on the DialogRE dataset, surpassing current state-of-the-art methods.

1. Introduction

Relation extraction aims to identify semantic relations between entities from unstructured text, supporting downstream tasks such as knowledge graph construction [1], search engines [2,3], and dialogue systems [4]. Traditional research has primarily focused on intra-sentence relation extraction, which involves identifying semantic associations between two entities within a single sentence. Significant progress has been made in this area, with the development of numerous high-performance models [5–8]. In recent years, some scholars have focused on document-level and dialogue-level relation extraction to adapt to more complex scenarios. Yao et al. [9] pioneered cross-sentential relation extraction by constructing DocRED,¹ the first document-level relation extraction dataset derived from Wikipedia, which effectively transcended conventional sentence-boundary constraints through interpropositional dependency modelling. Yu et al. [10] further advanced the computational linguistics community by manually annotating dialogues from the seminal American sitcom Friends, thereby establishing DialogRE,² the first benchmark dataset explicitly designed for Dialogue Relation Extraction (DRE). This groundbreaking work formally institutionalized DRE as an independent research paradigm within discourse understanding.

DRE aims to identify the semantic relation between given argument pairs, which are typically named entities, such as person names, place names, and other similar terms. As illustrated in Fig. 1, in multi-turn dialogues, given a dialogue text that includes speakers and their utterances, along with a pair of arguments, the goal of DRE is to determine the relations between the two arguments based on the dialogue context. In this scenario, trigger words denote lexical elements or phrasal constructs that provide crucial evidence for relation type identification. For instance, the phrase 'broke up' serves as the primary linguistic cue establishing a romantic relationship between Speaker 1 and Donna. However, analysis of the DialogRE dataset reveals that approximately 37% of instances lack such explicit indicators, instead exhibiting implicit relational patterns. The relationship between Speaker 2 and "sweetheart" exemplifies this challenge, wherein the semantic association must be inferred without direct lexical evidence. Consequently, the DRE task encounters three major challenges: (1) Coreference resolution difficulty due to dynamic changes in speaker identities and ambiguous pronoun references; (2) Modelling long-distance dependencies, as key semantic clues are dispersed across multiple turns of dialogue; (3) The sparsity of dialogue information and the redundancy of casual chatter introduce noise into the task of inter-entity relation recognition.

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Speaker	Utterance	
Speaker 1	Ugh, I cannot wait to ask her out!	
Speaker 2	Wait a minute...Joey. Joey you can't ask her out, she's your roommate. It-it'll be way too complicated.	
Speaker 3	Yeah, yeah man don't do it. I mean if you date her, then-then-then I can't date her.	
Speaker 4	All right, think about it. Now remember when you were going out with that girl Donna and you guys broke up. Remember how horrible it was when you guys bumped into each other at the supermarket?	
Speaker 1	Oh God, yeah.	
Speaker 4	Now imagine you live at the supermarket.	
Speaker 1	Okay!	
Speaker 4	No-no	
Speaker 1	Oh, Oh, you're right! I don't want that. I can't date her!	
Speaker 2	Yeah and you better watch the flirting too, cause you know, in such close quarters, it could be trouble.	
Speaker 1	Well, that's gonna be tough Mon. I mean it's hard for me to be around an attractive woman and not flirt.	
Speaker 2	Hmm, well you're around me all the time and you don't flirt.	
Speaker 1	You a little sad about that sweetheart?	
Argument Pair	Trigger	Relation
(Speaker 1, Donna)	broke up	per:girl/boyfriend
(Speaker 2, sweetheart)	-	per:alternate_names

Fig. 1. An example of dialogue-based relation extraction from the DialogRE dataset.

To address the limitations of existing methods in modelling semantic coherence across turns and understanding interactive behaviours. The Multi-Perspective Contextual Interaction Attention Network (MPCIA-Net) is proposed for DRE. The dialogue text is concatenated with the given pair of arguments as input to the pretrained language model, thereby obtaining context-aware initial representations. Through the Dual-stream Context Disentanglement Module, speaker perception features are systematically disentangled from utterance perception features. In the speaker perception channel, the individual delivering the utterance mitigates the dispersion of semantic information that arises from dynamic changes in their identity throughout the dialogue. Conversely, in the utterance perception channel, long-distance dependencies are effectively addressed through cross-turn dialogue information. This dynamic fusion of fine-grained features significantly enhances the model's capacity to decode complex interactive semantics. Then, the Dual-Attention Fusion Module establishes implicit mappings between speaker roles and semantic focus by leveraging attention interactions from both speaker-aware and utterance-aware perspectives. This approach not only explicitly models long-distance dependencies but also effectively resolves ambiguities in pronoun references, such as the cross-turn resolution of pronouns like "he/she".

The primary contributions of our work are outlined as follows:

- The Multi-Perspective Contextual Interaction Attention Network (MPCIA-Net) is proposed to effectively address complex dialogue structures in multi-turn dialogues and extract inter-entity relations across turns, a challenge that traditional methods often fail to tackle.
- The Dual-stream Context Disentanglement Module (DCDM) is introduced to enhance the modelling of multi-party dialogue structures and strengthen the model's ability to decode complex interactive semantics.
- Experimental results on the DialogRE dataset demonstrate that the MPCIA-Net achieved an F_1 score of 68.70%, significantly outperforming existing baseline models.

2. Related work

Relation extraction has made significant progress over the past few years, with early research primarily focused on intra-sentence relation extraction. The CNN-based method [18] captures the positional features between entities through local windows. The RNN architecture [19] utilizes sequence modelling to learn contextual dependencies, while attention mechanisms [20] enhance the representation of key semantic units through dynamic weight allocation. Although these methods perform well in intra-sentence relation extraction tasks, their fixed window modelling and short-range dependency capturing mechanisms struggle to handle the complex interactions across turns in dialogue scenarios. Current research methods for dialogue relation extraction can be primarily categorized into three types: sequence-based, graph-based, and trigger-word-enhanced approaches.

2.1. Sequence-based dialogue relation extraction

Sequence-based models employ LSTM [21] or BERT [10] to encode dialogues into long sequences, utilizing a series of complex mechanisms that include attention mechanisms and gating mechanisms to capture key information within the dialogue. These models effectively process the sequential data, allowing them to discern and emphasize important relational cues embedded across the conversation. Joshi et al. [22] propose a dialogue-aware BERT variant, which employs a Dynamic Span Masking strategy to randomly mask contiguous segments of utterances, thereby adapting it to dialogue settings. Xue et al. [11] adopt a model architecture perspective to further optimize and enhance the discriminative power of relation representations, proposing a Relation Refinement Gate (RRG) that adaptively extracts semantic representations for specific relations. However, sequence-level flattening results in the loss of dialogue structure information—merely concatenating multi-turn dialogues into a long sequence overlooks the unique structural characteristics of dialogue.

Table 1

Comparison of existing dialogue relation extraction methods with our proposed MPCIA-Net.

Year	Model	Advantage	Limitations
2022	SimplERE [11]	Adaptively extracts relation-specific semantic representations	Sequential flattening leads to loss of dialogue structure information
2023	DHGAT [12]	Models dialogue elements including utterances, words, speakers, arguments, and entity types as heterogeneous graph nodes; utilizes heterogeneous graph attention networks to model cross-sentence relations	Fixed graph structure, inability to dynamically adapt to different dialogue characteristics
2020	GDPNet [13]	Proposes DTWPool graph pooling method, refining latent multi-views learned from text sequences	Complex graph construction and high computational overhead
2021	TUCORE [14]	Introduces heterogeneous dialogue graph to model interactions among dialogue elements (e.g., speakers, turns, arguments)	Only focuses on turn-level interactions; neglects fine-grained word-level dependencies
2021	TWP [15]	First application of dual-layer GCN network for dialogue relation extraction	Fails to provide trigger features to the relation classification module
2022	TREND [16]	Connects triggers with dialogue context through vanilla attention mechanism	Lacks modelling of speaker roles
2023	TLAG [17]	Employs adaptive trigger fusion to reduce noise	Cannot capture cross-turn dependencies in dialogue context
Ours	MPCIA-Net	Decouples dialogue structure features; performs orthogonal projection of speaker and utterance information; eliminates semantic confusion	

2.2. Graph-based dialogue relation extraction

With the advancement of graph neural networks [23], graph-based dialogue relation extraction models structurally model the interactive relations among dialogue elements. They use nodes to represent tokens, utterances, or entities in a given dialogue and construct a graph by linking these nodes in various ways. Subsequently, they leverage the message-passing mechanism of graph neural networks to aggregate information. Chen et al. [12] employ a graph attention network to model information on utterances, words, speakers, arguments, and entity types, utilizing the message-passing mechanism of a heterogeneous graph attention network to capture cross-sentence relations within the dialogue. Xue et al. [13] propose a Gaussian Graph Generator (GG) that initializes the edges of latent multi-views by measuring the KL divergence between different Gaussian distributions, and introduce a graph pooling method, DTWPool, to refine the multi-view representations learned from the text. Lee and Choi [14] take into account the contextual representations of turn-level nodes within the dialogue and employs a graph convolutional network to propagate interactive information among these nodes. Zhang et al. [24] introduce a dual graph structure consisting of a dialogue hypergraph and a dialogue heterogeneous graph to capture dialogue contextual information, where the dialogue hypergraph leverages hyperedges to connect similar nodes, while the dialogue heterogeneous graph links different types of nodes and edges, thereby improving the representation of complex relations among multiple participants, and ultimately uses a graph convolutional network [25] to extract contextual and interactive information from the dialogue text.

2.3. Trigger-word-enhanced dialogue relation extraction

Recent studies have leveraged trigger word information to facilitate dialogue relation extraction. Zhao et al. [15] add two auxiliary tasks to enhance relation extraction, namely a speaker prediction task to identify speaker characteristics and a trigger word prediction task to provide contextual features for relation identification. Lin et al. [16] introduce a model called TREND, which possesses the capacity to generalize triggers. Through an attention mechanism, the model explicitly integrates trigger features and transfers them to unseen relations, thereby enhancing the interpretability of relation prediction. An et al. [17] enhance the performance of the trigger extractor through adaptive trigger fusion and the introduction of label-aware knowledge guidance, thereby reducing noise in relation extraction. Despite achieving favourable experimental results, the aforementioned methods do not fully leverage the inherent features of dialogue text and thus are unable to capture cross-turn dependencies within the dialogue completely.

Table 1 summarizes the key innovations and limitations of existing dialogue relation extraction methods. The proposed MPCIA-Net addresses these fundamental challenges by employing orthogonal subspace decomposition, which explicitly separates speaker identity from utterance content, and attention-based fusion mechanisms that capture the unidirectional semantic influence.

3. Method

This section first formulates the dialogue-based relation extraction task and then presents a detailed description of the proposed MPCIA-Net architecture, as shown in Fig. 2.

3.1. Task definition

Given a dialogue text D containing N turns, defined as $D = \{(s_1 : u_1), (s_2 : u_2), \dots, (s_N : u_N)\}$ and a set of argument pairs $A = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$, where the subject x_j and the object y_j are entities mentioned in the dialogue D . The objective of DRE is to determine the relation r from a predefined set of relations R for a given argument pair (x_j, y_j) . The descriptions of the corresponding symbols are provided in Table 2.

3.2. Input

In line with previous studies, we employ the pre-trained language model BERT³ [26] as an encoder to generate contextualized representations for dialogue and argument pairs. We apply the data preprocessing method developed by Yu et al. [10] to process the dialogue text. Given the dialogue $D = \{(s_1 : u_1), (s_2 : u_2), \dots, (s_N : u_N)\}$ and argument pair $a = (x_j, y_j)$, where s_i and u_i represent the speaker ID and corresponding dialogue text for the turn i , to help the model better capture the association between the dialogue text and the argument pair, we construct $\hat{D} = \{(\hat{s}_1, u_1), (\hat{s}_2, u_2), \dots, (\hat{s}_N, u_N)\}$. See Eq. (1).

$$\hat{s}_i = \begin{cases} [S_1], & \text{if } s_i = x_j, \\ [S_2], & \text{if } s_i = y_j, \\ s_i, & \text{else.} \end{cases} \quad (1)$$

Here, $[S_1]$ and $[S_2]$ are two special tokens. Additionally, we define $\hat{a} = (\hat{x}_j, \hat{y}_j)$, as shown in Eq. (2).

$$(\hat{x}_j, \hat{y}_j) = \begin{cases} ([S_1], y_j), & \text{if } s_i = x_j, \\ (x_j, [S_2]), & \text{if } s_i = y_j, \\ (x_j, y_j), & \text{else.} \end{cases} \quad (2)$$

Finally, we concatenate the dialogue text $\hat{D}$ and the argument pair $\hat{a}$ into a sequence as input for the pre-trained language model.

$$input = [CLS] + \hat{D} + [SEP] + \hat{x}_j + [CLS] + \hat{y}_j \quad (3)$$

Here, the first [CLS] token is used to learn the encoding that represents the global context of the dialogue, denoted as h^{cls1} , while the second [CLS] token is used to learn the association between the argument pair (x_j, y_j) and serves as its encoded representation h^{cls2} . The [SEP] token is used as a separator.

³ <https://huggingface.co/google-bert/bert-base-uncased>.

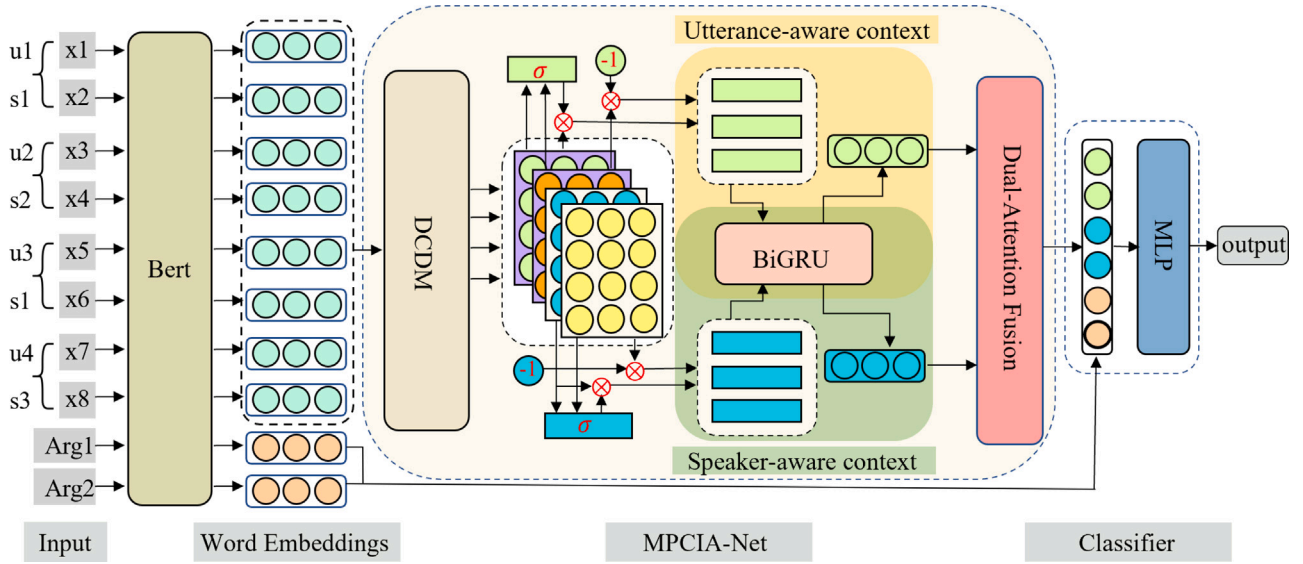


Fig. 2. The overall architecture of MPCIA-Net.

3.3. MPCIA-Net

In this paper, we introduce MPCIA-Net, which is illustrated in Fig. 2. This network architecture includes a Dual-stream Context Disentanglement Module and a Dual-Attention Fusion Module, which we will detail further below.

3.3.1. Dual-stream context disentanglement module

In the DRE task, the relations between entities exhibit speaker sensitivity, presenting a significant challenge in disentangling speaker-specific embeddings from the unified contextual representations generated by BERT. Meanwhile, to effectively capture the implicit coreference relations and semantic continuity inherent in utterances, we propose a Dual-stream Contextual Disentanglement Module, which employs dual-channel encoding to facilitate structured representation disentanglement.

From the dialogue text $\hat{D}$, we obtain the sequence of speakers $s = (s_1, s_2, \dots, s_N)$ and the boundary positions for each dialogue turn $t = (\tau_1, \tau_2, \dots, \tau_N)$, where each boundary is defined as $\tau_k = (u_{k,start}, u_{k,end})$, $k \in 1 \sim N$. In dialogue texts, speakers and utterance contents are highly coupled, and failing to distinguish the utterance content from its speaker clearly may lead to semantic confusion.

We decompose the global representation into orthogonal subspaces related to the speaker and the utterance content using parameter-isolated projection matrices, as described in Eq. (4).

$$\begin{aligned} h_s &= \text{LayerNorm}(h^{cls1} \cdot W_s + b_s), \\ h_u &= \text{LayerNorm}(h^{cls1} \cdot W_u + b_u), \\ W_s^T W_u &= 0 \end{aligned} \quad (4)$$

Here, W_s , b_s , W_u , and b_u are learnable parameters. To ensure feature independence between speaker identity and utterance content, the orthogonality constraint ($W_s^T W_u = 0$) is enforced through a carefully designed training mechanism: First, a regularization term $\ell_{\text{orth}} = \lambda \|W_s^T W_u\|_F^2$ is incorporated into the loss function with an adaptive weight λ that adjusts based on the orthogonality violation degree. Second, SVD-based hard orthogonalization is performed periodically (every 100 steps) to guarantee strict orthogonality, where the projection matrices are decomposed and reconstructed as $W_s = U_{:,d} \sqrt{\Sigma_1}$ and $W_u = U_{:,d} \sqrt{\Sigma_2}$ from $[U, \Sigma, V] = \text{SVD}([W_s, W_u])$. This dual-mechanism approach combines soft gradient-based optimization with periodic hard constraints. This combination ensures robust orthogonal

Table 2

Notation and symbol definitions.

Symbol	Description
Input and dialogue components	
D	Dialogue text containing N turns
N	Number of dialogue turns
s_i	Speaker ID for turn i
u_i	Utterance text for turn i
A	Set of argument pairs
(x_j, y_j)	Subject-object entity pair
R	Predefined set of relations
r	Target relation to be extracted
$[S_1], [S_2]$	Special tokens for marking entities
Representations and embeddings	
h^{cls1}	Global dialogue context encoding
h^{cls2}	Argument pair encoding
h_s	Speaker-aware representation
h_u	Utterance-aware representation
h^*	Final fused representation
$\tilde{h}_s, \tilde{h}_u$	Context representation sequences
h_s^*, h_u^*	Final BiGRU hidden states
h_s^{intra}, h_s^{inter}	Intra/inter-speaker features
h_u^{intra}, h_u^{inter}	Intra/inter-utterance features
Model parameters	
W_s, W_u	Orthogonal projection matrices $\in \mathbb{R}^{d \times d}$
b_s, b_u	Bias vectors $\in \mathbb{R}^d$
W_q, W_k, W_v	Attention weight matrices
W_{start}, W_{end}	Trigger span prediction weights
d	Hidden dimension size
Masking and attention	
$M^{intra-spk}$	Intra-speaker attention mask
$M^{inter-spk}$	Inter-speaker attention mask
$M^{intra-utt}$	Intra-utterance attention mask
$M^{inter-utt}$	Inter-utterance attention mask
τ_k	Boundary positions for dialogue turn k
p_s, p_u	Gating weights for feature fusion
Loss functions	
$\mathcal{L}_r$	Relation classification loss
$\mathcal{L}_b$	Binary trigger prediction loss
$\mathcal{L}_t$	Trigger span extraction loss
ℓ_{orth}	Regularization loss
$\mathcal{L}$	Total loss function

Algorithm 1: Orthogonal Subspace Decomposition

Input: Context $\mathbf{h}^{\text{cls}}$, orthogonality weight λ , thresholds τ
Output: $\mathbf{W}_s^T \mathbf{W}_u = 0$ (orthogonal projections)
Initialize: $\mathbf{W}_s, \mathbf{W}_u \in \mathbb{R}^{d \times d}$, $\mathbf{b}_s, \mathbf{b}_u \in \mathbb{R}^d$;
 $\lambda \leftarrow 0.01$, $k \leftarrow 100$ // orthogonality weight, SVD interval
for each epoch do
 for each batch $\mathbf{H}_b$ in $\mathbf{h}^{\text{cls}}$ do
 $\mathbf{h}_s \leftarrow \gamma_s \odot \text{Norm}(\mathbf{H}_b \mathbf{W}_s + \mathbf{b}_s) + \beta_s$;
 $\mathbf{h}_u \leftarrow \gamma_u \odot \text{Norm}(\mathbf{H}_b \mathbf{W}_u + \mathbf{b}_u) + \beta_u$;
 $\ell_{\text{orth}} \leftarrow \lambda \|\mathbf{W}_s^T \mathbf{W}_u\|_F^2$;
 $\ell_{\text{task}} \leftarrow \text{CE}(f(\mathbf{h}_s \oplus \mathbf{h}_u), y)$;
 $\ell \leftarrow \ell_{\text{task}} + \ell_{\text{orth}}$;
 // Periodic SVD orthogonalization
 if $\text{step} \% k = 0$ then
 $[\mathbf{U}, \Sigma, \mathbf{V}] \leftarrow \text{SVD}(\mathbf{W}_s, \mathbf{W}_u)$;
 $\mathbf{W}_s \leftarrow \mathbf{U}[:, :d] \sqrt{\Sigma_1}$, $\mathbf{W}_u \leftarrow \mathbf{U}[:, d:] \sqrt{\Sigma_2}$;
 end
 // Adaptive λ adjustment
 $\lambda \leftarrow \begin{cases} \lambda \times 1.2 & \text{if } \|\mathbf{W}_s^T \mathbf{W}_u\|_F^2 > \tau_h; \\ \lambda \times 0.9 & \text{otherwise} \end{cases}$;
 // Gradient update
 $\mathbf{W}_s \leftarrow \mathbf{W}_s - \alpha \nabla_{\mathbf{W}_s} \ell$;
 $\mathbf{W}_u \leftarrow \mathbf{W}_u - \alpha \nabla_{\mathbf{W}_u} \ell$;
 end
end

subspace separation, thereby minimizing mutual information between speaker and utterance features. The complete orthogonal subspace decomposition procedure is formalized in Algorithm 1, ensuring both clarity and reproducibility.

Within the speaker stream h_s and the utterance stream h_u , we further decompose the features into intra-speaker and inter-speaker characteristics, as well as intra-utterance and inter-utterance attributes, utilizing an asymmetric attention mechanism. Therefore, in accordance with Eq. (5), we have developed four masking matrices to regulate the direction of decoupling through the attention mechanism.

$$\begin{aligned}
 M_{i,j}^{\text{intra-sp}} &= \begin{cases} 1, & \text{if } s_i = s_j \\ 0, & \text{otherwise} \end{cases} \\
 M_{i,j}^{\text{inter-sp}} &= \begin{cases} 1, & \text{if } s_i \neq s_j \\ 0, & \text{otherwise} \end{cases} \\
 M_{i,j}^{\text{intra-utt}} &= \begin{cases} 1, & \text{if } \tau_i = \tau_j \\ 0, & \text{otherwise} \end{cases} \\
 M_{i,j}^{\text{inter-utt}} &= \begin{cases} 1, & \text{if } \tau_i \neq \tau_j \\ 0, & \text{otherwise} \end{cases}
 \end{aligned} \tag{5}$$

The four types of masking matrices delineate the attention scope using distinct spatial connection patterns, as illustrated in Fig. 3. The speaker mask 3(a) exhibits a block diagonal pattern, indicating constraints applicable within the same speaker; meanwhile, the utterance mask 3(b) presents sparse inter-turn connections, whose intensity is modulated by semantic similarity.

Upon acquiring the semantic masking matrices, we implement a directed feature decoupling of the contextual representations encoded by BERT through a sophisticated multi-head attention mechanism. This process involves projecting the hidden states into designated subspaces using dynamically learnable attention weights, thereby facilitating meticulous and explicit segregation of speaker-aware features from utterance-aware features. This method ensures a refined differentiation,

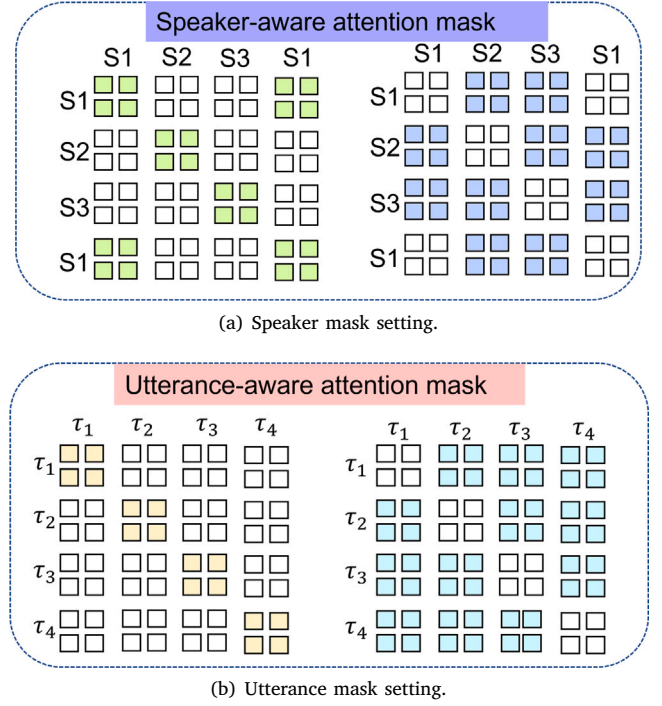


Fig. 3. Schematic diagram of the mask matrix.

pivotal for nuanced semantic analysis. See Eq. (6).

$$\begin{aligned}
 Q_\beta^\gamma &= h_\beta W_q^{\gamma-\beta} \\
 A_\beta^\gamma &= \text{Softmax} \left(\frac{Q_\beta^\gamma (h_\beta W_k^{\gamma-\beta})^T}{\sqrt{d_k}} \odot M^{\gamma-\beta} \right) \\
 h_\beta^\gamma &= A_\beta^\gamma h_\beta W_v^{\gamma-\beta} \\
 \beta &\in (\text{spk}, \text{utt}) \\
 \gamma &\in (\text{intra}, \text{inter})
 \end{aligned} \tag{6}$$

We control the direction of attention through the parameters β and γ , enabling the acquisition of perceptual features at varying granularities.

3.3.2. Dual perception context vector representation

To effectively integrate the complementary information present within each channel, we utilize a sophisticated gating mechanism that computes the retention ratios of channel-specific information. This mechanism is meticulously defined in Eq. (7), ensuring precise modulation of information flow across the channels. In this configuration, E is derived from the raw contextual representation h^{cls} of the dialogue, as generated by the BERT. The features C_1 and C_2 are specific to individual channels. By employing subtraction, we identify the discrepancies between the channel features and the original representation, using these differences to refine the overarching global information with nuanced local insights. Furthermore, the dot product operation is utilized as a metric to gauge the similarity between the channel features and the original representation, thereby enhancing the semantic representation.

$$\begin{aligned}
 E_1^{(1)} &= \text{ReLU} (FC^{(1)}([E, C_1, E - C_1, E \odot C_1])) \\
 E_1^{(2)} &= \text{ReLU} (FC^{(2)}([E, C_2, E - C_2, E \odot C_2])) \\
 G_{(E, C_1, C_2)} &= \text{Sigmoid}(FC^{(3)}([E_1^{(1)}, E_1^{(2)}]))
 \end{aligned} \tag{7}$$

Having ascertained the information retention ratios for each channel within the speaker-aware and utterance-aware contexts, we proceed

to construct the contextual representation sequences $\tilde{h}_s$ and $\tilde{h}_u$, as delineated in Eq. (8). This process ensures that the sequences are precisely tailored to capture the distinct nuances of the respective contextual domains.

$$\begin{aligned} p_s &= G_1(E, h_s^{intra}, h_s^{inter}) \\ p_u &= G_2(E, h_u^{intra}, h_u^{inter}) \\ \tilde{h}_s &= p_s \odot h_s^{intra} + (1 - p_s) \odot h_s^{inter} \\ \tilde{h}_u &= p_u \odot h_u^{intra} + (1 - p_u) \odot h_u^{inter} \end{aligned} \quad (8)$$

It is crucial to emphasize that, to guarantee the absence of coupling between speaker-aware and utterance-aware information, the dual-channel linear layer, denoted as FC, utilizes two distinct sets of parameters, ensuring independent processing within each channel.

To acquire channel-aware dialogue information representations, the contextual representation sequences $\tilde{h}_s$ and $\tilde{h}_u$ are fed into a BiGRU (Bidirectional Gated Recurrent Unit). Assuming the hidden states of the BiGRU are $(h_1, h_2, \dots, h_k)$, for $\forall j \in (1 \sim k)$.

$$\begin{aligned} \bar{h}_j &= \overline{\text{GRU}}(\bar{h}_{j-1}, \bar{L}[j]) \\ \tilde{h}_j &= \overline{\text{GRU}}(\tilde{h}_{j-1}, \tilde{L}[j]) \\ h_j &= [\bar{h}_j; \tilde{h}_j] \end{aligned} \quad (9)$$

To capture speaker context-aware and utterance context-aware information, we employ Eq. (9), which designates the final hidden states of the BiGRU as the definitive contextual information vectors, h_s^* and h_u^* . We will subsequently integrate these vectors using an advanced attention mechanism to enhance the contextual alignment between them.

3.3.3. Dual-attention fusion

Given the unidirectional influence of speaker identity on utterance semantics within dialogue texts, we utilize an asymmetric attention mechanism to effectively integrate the speaker context-aware information h_s^* with the utterance context-aware information h_u^* . This approach ensures a nuanced fusion that respects the directional impact of speaker characteristics on the conveyed meanings.

$$\begin{aligned} \alpha &= \text{Softmax} \left(\frac{h_u^* W_q (h_s^* W_k)^T}{\sqrt{d_k}} \right) \\ h^* &= \text{LayerNorm}(W_v h_s^* \alpha + h_u^*) \end{aligned} \quad (10)$$

As delineated in Eq. (10), the asymmetric attention mechanism adeptly models the unidirectional semantic dependency from the speaker to the utterance, effectively mitigating semantic confusion caused by the speaker's influence. This strategic modelling ensures clearer and more precise semantic interpretation within dialogues. In this framework, α denotes the modulation weights assigned by the speaker role to various semantic units in the utterance content. By incorporating residual connections within the normalization layer, the model not only preserves the integrity of the original utterance but also facilitates the seamless integration of speaker information into the utterance, enhancing the overall semantic coherence.

3.4. Classifier

After processing through the MPCIA-Net, we obtain the final representation of the dialogue text, denoted as h^* . This representation is concatenated with the encoded representations of entities x_j and y_j , labelled as h^{cls2} , and then forwarded to the classification layer. We decompose the DRE task into two subtasks: trigger word prediction and relation classification. We employ a softmax classifier to predict the labels of entities x_j and y_j from a predefined set of relations R . As outlined in Eq. (11), given the dialogue D and entities x_j, y_j , we

concatenate h^* and h^{cls2} to form the input to the softmax classifier. This classifier is then trained using cross-entropy loss.

$$\begin{aligned} p(r_i | (D, x_j, y_j)) &= \text{Softmax}(w_h [h^*; h^{cls2}] + b_h) \\ r' &= \text{argmax}_p(r_i | (D, x_j, y_j)) \\ \mathcal{L}_r &= \text{CELoss}(r, r') \end{aligned} \quad (11)$$

Trigger word prediction is a binary classification task that utilizes a sigmoid function as the classifier, with binary cross-entropy loss serving as the loss function, as specified in Eq. (12).

$$\begin{aligned} p(t | (D, x_j, y_j)) &= \text{Sigmoid}(w_t [h^*; h^{cls2}] + b_t), \\ \mathcal{L}_b &= \text{BCELoss}(p(t | (D, x_j, y_j)), t) \end{aligned} \quad (12)$$

If a trigger word exists between entities, following the approach of [16], we incorporate a trigger word span prediction task to aid in relation extraction, as detailed in Eq. (13). We predict the start and end pointers of the trigger word.

$$\begin{aligned} h^S &= W^{start} h^* + b^{start}, \\ h^E &= W^{end} h^* + b^{end}, \end{aligned} \quad (13)$$

We optimize the trigger span extraction task using cross-entropy loss.

$$\begin{aligned} p(\text{start} = i, \text{end} = j) &= \text{Softmax}(h_i^S h_j^E) \\ \mathcal{L}_t &= \text{CELoss}(\text{start}, \text{start}') + \text{CELoss}(\text{end}, \text{end}') \end{aligned} \quad (14)$$

Finally, the optimization objective is to minimize the loss function $\mathcal{L}$, as defined in Eq. (15).

$$\mathcal{L} = \mathcal{L}_r + \mathcal{L}_b + \mathcal{L}_t + \ell_{\text{orth}} \quad (15)$$

where ℓ_{orth} is the loss of regularization.

Algorithm 2: MPCIA-Net Training Process

Input: Training data D_{train} , Validation data D_{val}

Output: Trained model parameters θ

Initialize: BERT encoder, projection matrices W_s, W_u , classifier weights;

for epoch = 1 **to** max_epochs **do**

for batch in D_{train} **do**

 // Forward pass

$h^{cls1}, h^{cls2} \leftarrow \text{BERT_encode}(\text{batch});$

$h_s, h_u \leftarrow \text{orthogonal_decomposition}(h^{cls1});$

$h^* \leftarrow \text{dual_attention_fusion}(h_s, h_u);$

 // Compute losses

$\mathcal{L}_r \leftarrow \text{relation_classification_loss}(h^*, h^{cls2});$

$\mathcal{L}_b \leftarrow \text{binary_trigger_loss}(h^*, h^{cls2});$

$\mathcal{L}_t \leftarrow \text{trigger_span_loss}(h^*, h^{cls2});$

$\mathcal{L}_{\text{orth}} \leftarrow \lambda \|W_s^T W_u\|_F^2;$

$\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_r + \mathcal{L}_b + \mathcal{L}_t + \mathcal{L}_{\text{orth}};$

 // Backward pass and optimization

$\theta \leftarrow \theta - \alpha \nabla_{\theta} \mathcal{L}_{\text{total}};$

 // Periodic orthogonalization (every 100 steps)

if step % 100 = 0 **then**

$W_s, W_u \leftarrow \text{apply_SVD_orthogonalization}(W_s, W_u);$

end

end

 // Validation

$F1_{\text{val}} \leftarrow \text{evaluate}(\text{model}, D_{\text{val}});$

if $F1_{\text{val}} > \text{best_F1}$ **then**

 save_model(θ);

end

else if no_improvement_count > patience **then**

 break;

 // Early stopping

end

end

Table 3
Statistics of the DialogRE dataset.

DialogRE	Train	Dev	Test
# Conversations	1073	358	357
Average dialogue length	229.5	224.1	214.2
# Argument pairs	5963	1928	1958
Average turns	13.1	13.1	12.4
Average speakers	3.3	3.2	3.3

3.5. Train and inference

The training process of MPCIA-Net follows an end-to-end multi-task learning paradigm, as depicted in Algorithm 2. The model optimizes three classification objectives simultaneously: relation classification loss $\mathcal{L}_r$, trigger detection loss $\mathcal{L}_b$, and trigger span prediction loss $\mathcal{L}_t$. Additionally, an orthogonality regularization term $\mathcal{L}_{\text{orth}} = \lambda \|\mathbf{W}_s^T \mathbf{W}_u\|_F^2$ ensures feature independence between speaker and utterance representations.

During inference. Given a test dialogue and entity pair, the model performs a single forward pass through the trained network, utilizing the optimized orthogonal projection matrices that satisfy $\mathbf{W}_s^T \mathbf{W}_u \approx 0$. The final prediction is obtained by selecting the relation with maximum probability from the 36-class softmax output.

4. Experiments

In this section, we will introduce the dataset and several baseline comparisons.

4.1. Datasets

The experimental analysis employed the DialogRE dataset,⁴ which is the first manually annotated resource specifically tailored for Dialogue Relation Extraction (DRE). Derived from the daily interactions of the main characters in the American television series “Friends”, it encompasses 36 distinct relation types, 1788 dialogues, and 8119 argument pairs. Comprehensive statistics for the DialogRE dataset are presented in Table 3.

In realistic dialogue scenarios, information continuously emerges as conversations unfold. Consequently, we have introduced a novel evaluation metric, $F1_c$, within the DialogRE dataset to assess model performance under conditions of dynamically updating information. Furthermore, we present the $F1$ scores of our model, MPCIA-Net, in comparison with all baseline models on the DialogRE dataset, thereby providing a comprehensive analysis of their respective performances.

4.2. Baseline models

To assess MPCIA-Net’s performance comprehensively, we compare it with four categories of representative baseline models on the DialogRE dataset.

Sentence-level models are primarily designed for relation extraction within individual sentences. We have chosen three prototypical methods—CNN [27], LSTM [28], and BiLSTM [29]. These models represent the most conventional approaches in the domain of sentence-level relation extraction, providing a clear benchmark for evaluating the effectiveness of more advanced methodologies.

Sequence-based models process dialogue content and arguments by simply concatenating them as input for PLMs. This category includes the most common models such as BERT [10] and RoBERTa [30]; BERT [10] employs special tokens to explicitly differentiate speakers,

thereby enhancing the model’s ability to capture speaker-specific relation; SpanBERT [22] utilizes a Dynamic Span Masking strategy to randomly mask continuous segments of speech, improving the representational power of the model; SimpleRE [11] adaptively extracts semantic representations specific to relation.

Graph-based models utilize graph neural networks to model relations among entities, utterances, and speakers, facilitating the transmission of interaction information through the network. DHGAT [12] constructs a dialogue heterogeneous graph and propagates information via a graph attention network; LSR [31] introduces latent structural relations and employs graph convolutional networks for relational inference; GDPNet [13] leverages a Gaussian Graph Generator to build multi-view representations that refine dialogue features more effectively; TUCORE [14] constructs round-level context-aware graphs to extract relation in a manner that understands dialogues; HG2G [24] captures multi-level features by constructing dialogue hypergraphs and aggregating sentence information using heterogeneous graph convolutions.

Trigger-enhanced models leverage trigger word information to guide relation extraction. TWP [15] introduces trigger word prediction as an auxiliary task; TREND [16] proposes a trigger word span prediction task, utilizing trigger word information for relation extraction; TLAG [17] introduces a trigger word adaptive aggregation module, incorporating label-aware knowledge to guide relation extraction. However, these methods do not fully comprehend dialogues and are unable to handle relation extraction across dialogue turns.

4.3. Evaluate metric

To evaluate the model’s capability for timely relation extraction during dialogue, we adopt a novel evaluation metric $F1_c$ in dialogue assessment. This metric treats each utterance concatenated with all preceding utterances as text, and computes precision and recall for each turn d , where d represents the number of dialogue utterances. The $F1_c$ score for a single dialogue is the average of $F1_c$ scores across all turns. The corresponding P_c and R_c values are calculated as shown in Eqs. (16) to (19):

$$P_c^{(D,s,o)} = \frac{\sum_{i=1}^d |T_{\text{per}_i} \cap T_{\text{true}} \cap T_{\text{turn}_i}|}{\sum_{i=1}^d |T_{\text{per}_i} \cap T_{\text{turn}_i}|} \quad (16)$$

$$R_c^{(D,s,o)} = \frac{\sum_{i=1}^d |T_{\text{per}_i} \cap T_{\text{true}} \cap T_{\text{turn}_i}|}{\sum_{i=1}^d |T_{\text{true}} \cap T_{\text{turn}_i}|} \quad (17)$$

$$T_{\text{turn}_i} = \{r \mid i \geq \max\{J_s, J_o, J_t\}\} \quad (18)$$

$$F1_c = \frac{1}{P_c} + \frac{1}{R_c} \quad (19)$$

Here, s and o represent the subject entity and object entity, respectively. T_{per_i} indicates that the model can correctly predict the relation label between s and o at the i th utterance in the given dialogue D . T_{true} represents the ground truth relation label between s and o . J_s, J_o, J_t represent the indices of the utterances in which the subject entity, object entity, and trigger word make their initial appearance.

5. Result

5.1. Main result

In Table 4, we report the experimental results of MPCIA-Net and other baselines on the DialogRE dataset. Here, * denotes the locally reproduced results, which also serve as the comparative baselines in Fig. 5. Compared to the baselines, MPCIA-Net demonstrates a significant improvement in both $F1$ and $F1_c$ scores on the validation and test

⁴ <https://github.com/nlpdata/dialogre>.

Table 4
Experimental results on the DialogRE dataset.

Type	Model	DialogRE-Dev		DialogRE-Test	
		F_1 (%)	F_{1c} (%)	F_1 (%)	F_{1c} (%)
Sentence-level	CNN [27]	46.1	43.7	48.0	45.0
	LSTM [28]	46.7	44.2	47.4	44.9
	BiLSTM [29]	48.1	44.3	48.6	45.0
Sequence-based	BERT [10]	60.6	55.4	58.5	53.2
	BERTs [10]	63.0	57.3	61.2	55.4
	SpanBERT [22]	64.6	58.8	61.8	55.8
	RoBERTa [30]	65.2	61.4	62.8	58.8
	SimpleRE [11]	66.8	61.0	66.3	60.2
Graph-based	DHGAT [12]	60.2	57.1	59.9	55.8
	DHGAT ^a	59.3 ^a	55.4 ^a	56.6 ^a	54.7 ^a
	LSR [31]	62.8	58.4	61.4	56.2
	GDPNet [13]	67.1	61.5	64.9	60.1
	TUCORE [14]	66.8	61.0	65.5	60.2
	HG2G [24]	–	–	68.0	62.7
Trigger-enhanced	TWP [15]	66.8	61.05	65.5	60.5
	TREND ^a	64.3 ^a	61.8 ^a	64.8 ^a	60.9 ^a
	TREND [16]	66.8	61.5	65.5	60.5
	TLAG [17]	68.81	61.09	66.6	60.8
	MPCIA-Net(ours)	69.01	63.21	68.70	62.44

^a These results indicate our local reproduction of the corresponding models under identical experimental conditions for fair comparison.

Table 5
Ablation results of MPCIA-Net on DialogRE, w/o indicates the performance without that module.

Model	DialogRE-Dev		DialogRE-Test	
	F_1 (%)	F_{1c} (%)	F_1 (%)	F_{1c} (%)
MPCIA-Net _{w/o} DCDM	65.24	59.66	64.93	58.65
MPCIA-Net _{w/o} Utt-aware	64.95	59.37	65.54	59.12
MPCIA-Net _{w/o} spk-aware	66.16	60.04	65.80	59.39
MPCIA-Net _{w/o} Dual-att	64.76	59.01	64.81	59.00
MPCIA-Net	69.01	63.21	68.70	62.44

sets. On the test set, our F_1 and F_{1c} scores have reached 68.70% and 62.44%, respectively, substantiating the efficacy of our approach.

Compared to **sentence-level** models, our approach demonstrates significant advantages in dialogue scenarios, achieving substantial improvements in both F_1 and F_{1c} metrics. The inherent complexity of interactions within dialogues poses challenges for sentence-level inference, which struggles to capture the intricate relations present in conversations.

In comparison to **sequence-based** methods, our model continues to exhibit robust performance. The unique structure of dialogues leads to a flattened connection between dialogue texts and entities, inevitably resulting in the loss of structural information that can adversely affect performance.

When compared with **graph-based** methods, our approach maintains superior performance. Although graph-based approaches possess certain advantages in modelling dialogue structures, they still face limitations regarding information transmission due to the lower information density typically found in dialogues.

Finally, relative to **trigger-enhanced** methods, our approach retains distinct advantages. TWP [15] does not transmit trigger word features to the classification module; TREND [16] employs a simple classifier as a trigger word predictor, introducing noise into the relation extraction process; While TLAG [17] guides relation extraction through label-aware knowledge integration, it overlooks the inherent complexities of dialogues and does not fully capture dependency information across the dialogue context.

5.2. Ablation studies

In order to verify the effectiveness of each component in the MPCIA-Net model, we conducted ablation experiments on DialogRE. “-w/o”

indicates that the corresponding module was removed from MPCIA-Net. The results are shown in Table 5.

When the Dual-stream Context Decoupling Module (DCDM) is removed, the unified context representation generated by the BERT encoder is applied to the subsequent dual-channel perceptual features, leading to a drop of nearly 3 percentage points in F_{1c} on the test set. This demonstrates that separating speaker-aware and utterance-aware features with DCDM can effectively improve the model’s performance. Subsequently, we separately remove the utterance-perception and speaker-aware channels, and the model performance shows a significant decline, indicating that the dual-channel information is complementary. The utterance-aware channel focuses on modelling the text content and its semantic associations, while the speaker-aware channel focuses on the differences in information brought by the participants’ roles and their interaction patterns. The two channels working together can significantly enhance the model’s capability for understanding and predicting relations within dialogues. Finally, when we replaced the bidirectional attention fusion module with an operation that simply concatenates the two vectors and then applies average pooling, the performance also experienced a decline. Although the pooling operation can still integrate multi-channel information to some extent, it lacks the more fine-grained semantic alignment and interaction modelling capabilities provided by the bidirectional attention mechanism.

It is worth noting that the performance decreases for any module removed from MPCIA-Net compared to the most recent baseline TLAG.

5.3. Analysis of long-distance dependency extraction performance

Furthermore, to explore MPCIA-Net’s ability in modelling dialogue coherence, we compared and visualized its relation extraction performance with that of other models across different dialogue rounds.

As illustrated in Fig. 4, under ablation conditions, the F_1 and F_{1c} scores exhibit similar declining patterns. Using 5-round intervals as the analytical unit, when the DCDM module is removed (MPCIA-Net w/o DCDM), the model’s average F_1 score decreases from 68.84% to 66.85% in dialogues within the first 10 rounds, compared to MPCIA-Net’s decline from 70.38% to 67.49%, yielding a relative performance degradation ratio of 68%. For dialogues exceeding 15 rounds, the average F_1 score under ablation conditions plummets to 60.01%, while MPCIA-Net maintains 65.75%, representing a relative degradation of 146.03%—an exponential increase compared to shorter dialogues. Simultaneously, the F_{1c} score under ablation conditions drops from 65.51% to 63.57% in the first 10 rounds, and dramatically falls to 52.42% after 15 rounds, with a relative decline exceeding 1.70. This progressive deterioration pattern demonstrates that in extended dialogues, where speaker roles and contextual references become increasingly complex, the dual-stream architecture of DCDM proves crucial for maintaining semantic coherence.

Similarly, the removal of the DAF module (MPCIA-Net w/o Dual-att) results in a 1.13% F_1 score reduction in short dialogues, with a more substantial 5.17% performance drop observed in extended dialogues. As dialogue length increases, the importance of the dual-attention mechanism becomes increasingly evident, with its core function being the resolution of cross-turn dependencies and the maintenance of speaker-utterance alignment. The complete model consistently outperforms all ablated variants, with the increasing degradation in both F_1 and F_{1c} scores as dialogue complexity rises further validating the superiority of the proposed method in long-range modelling.

These empirical results validate two key hypotheses: (1) The complementary nature of the DCDM and DAF modules is confirmed, with each addressing distinct challenges in dialogue understanding—DCDM handles context disentanglement while DAF manages attention distribution across multiple perspectives; (2) The proposed architecture demonstrates robust scalability, as evidenced by its ability to maintain stable performance even with a 300% increase in dialogue length. The model’s superior performance in extended dialogues holds significant

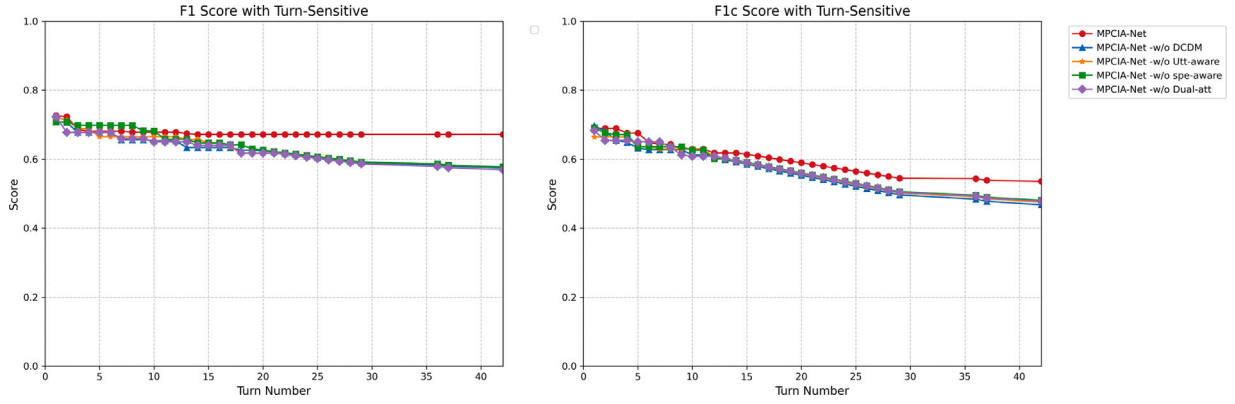


Fig. 4. Visualization results comparing the relation extraction performance of MPCIA-Net with versions where certain modules are removed, across each dialogue round.

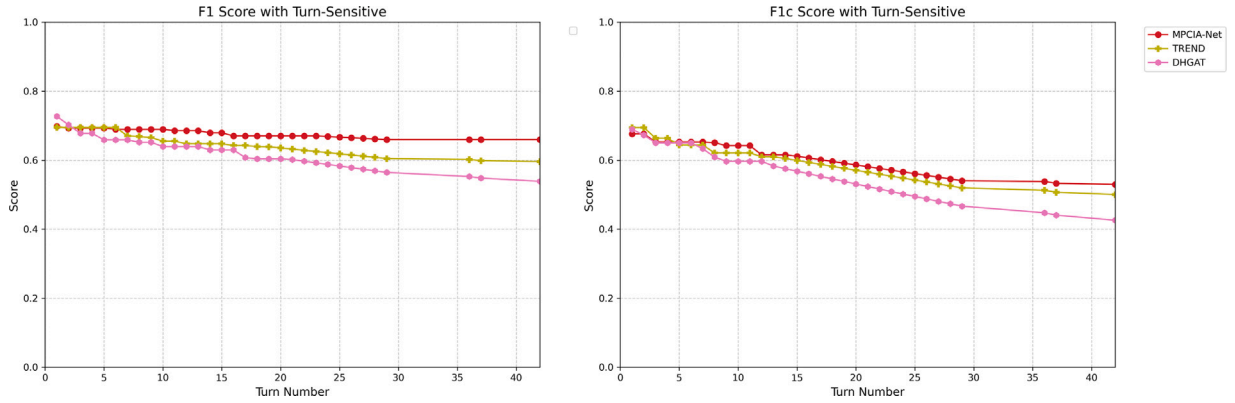


Fig. 5. Visualization results comparing the relation extraction performance of MPCIA-Net with the latest baseline across each dialogue round.

practical value, as real-world conversational scenarios typically involve multi-turn interactions with complex dynamic relationships among speakers.

Furthermore, a comparative analysis was conducted to evaluate the relation extraction capabilities of MPCIA-Net, TREND, and DHGAT in multi-turn dialogue scenarios. As illustrated in Fig. 5: The three models exhibit significantly different performance patterns across varying dialogue lengths. For dialogues with fewer than 15 rounds, the models demonstrate comparable performance: MPCIA-Net achieves an average F_1 score of 68.93%, marginally outperforming TREND at 67.82% and DHGAT at 66.74%. However, as dialogue length increases, a substantial performance divergence emerges. When dialogue rounds exceed 20, MPCIA-Net demonstrates clear architectural advantages, maintaining an F_1 score of 65.75%, while TREND and DHGAT decline to 61.79% and 58.69%, respectively. As dialogue complexity increases, traditional models encounter challenges with accumulating context noise and pronoun resolution ambiguity, whereas MPCIA-Net effectively addresses these challenges through its superior architectural design.

This widening performance gap can be attributed to two key factors: (1) The speaker perception channel in the Dynamic Adaptive Fusion (DAF) module effectively maintains speaker identity consistency throughout extended dialogues, preventing the semantic drift observed in baseline models; (2) The adaptive fusion mechanism dynamically adjusts the contribution weights of both channels based on dialogue complexity, providing robust support for handling variable conversational patterns.

6. Conclusion and future work

This paper addresses the challenges posed by dynamic interactions and cross-turn dependencies in dialogue relation extraction tasks by

proposing a Multi-Perspective Contextual Interaction Attention Network (MPCIA-Net). Experimental results on the DialogRE dataset show that MPCIA-Net outperforms existing baselines, yielding a 2% improvement in the F_1 score. Ablation studies confirm that the DCDM effectively decouples and integrates fine-grained contextual features of dialogues. The DAF module enhances the modelling of dialogue action coherence by integrating dual features of speaker perception and utterance perception through its attention mechanism. Furthermore, we visualized MPCIA-Net's relation extraction capability across varying dialogue turns. When the dialogue exceeds 15 turns, the superiority of our method becomes particularly evident, demonstrating a 1.70 improvement over baseline models, which further confirms MPCIA-Net's advantages in understanding long multi-turn dialogues. In future work, we will explore model compression techniques to enhance deployment efficiency. Recent advances in neural network quantization [32–36] provide promising directions for reducing model complexity while maintaining performance. Additionally, we will investigate context-irrelevant denoising methods to further improve relation extraction accuracy.

7. Limitations

While MPCIA-Net achieves state-of-the-art performance, its computational requirements present challenges for resource-constrained environments. The modular architecture of DCDM and DAF modules provides opportunities for targeted optimization using recent quantization techniques [32,34] and knowledge distillation methods [33,35]. Future iterations will focus on developing lightweight variants suitable for practical deployment [36].

CRedit authorship contribution statement

Shen Yatian: Writing – review & editing, Supervision, Resources.
Deng Guosong: Writing – review & editing, Writing – original draft, Validation, Methodology.
Qi Chaozhao: Investigation, Data curation.
Wang Songyang: Visualization, Validation.
Yanli Xu: Data curation.
Han Daojun: Supervision, Resources.
Lu Chengeng: Formal analysis.

Code availability

Our code is available at <https://anonymous.4open.science/r/MPCI-A-Net-3939/>

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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